Tasks on ECS

1. Setup a highly available ecs cluster with load balancer and dynamic port mapping.

NOTE:

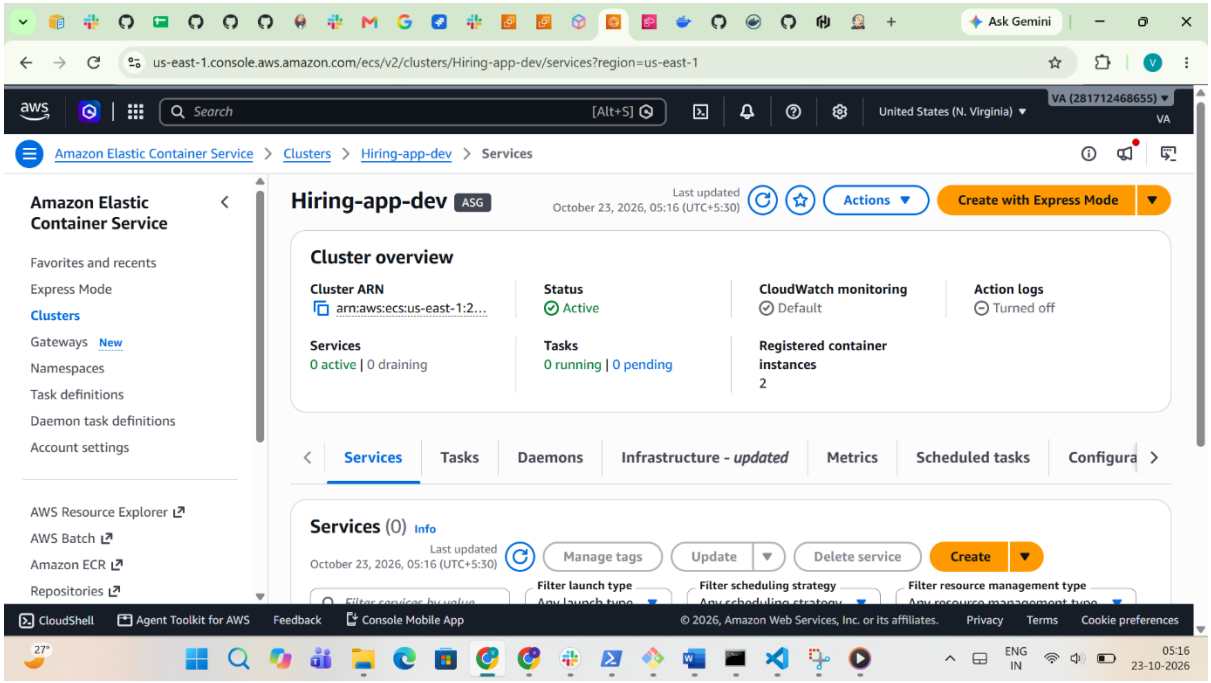
Your cluster should maintain atleast 6 tasks at any point of time and should be highly availble across multiple AZ's.

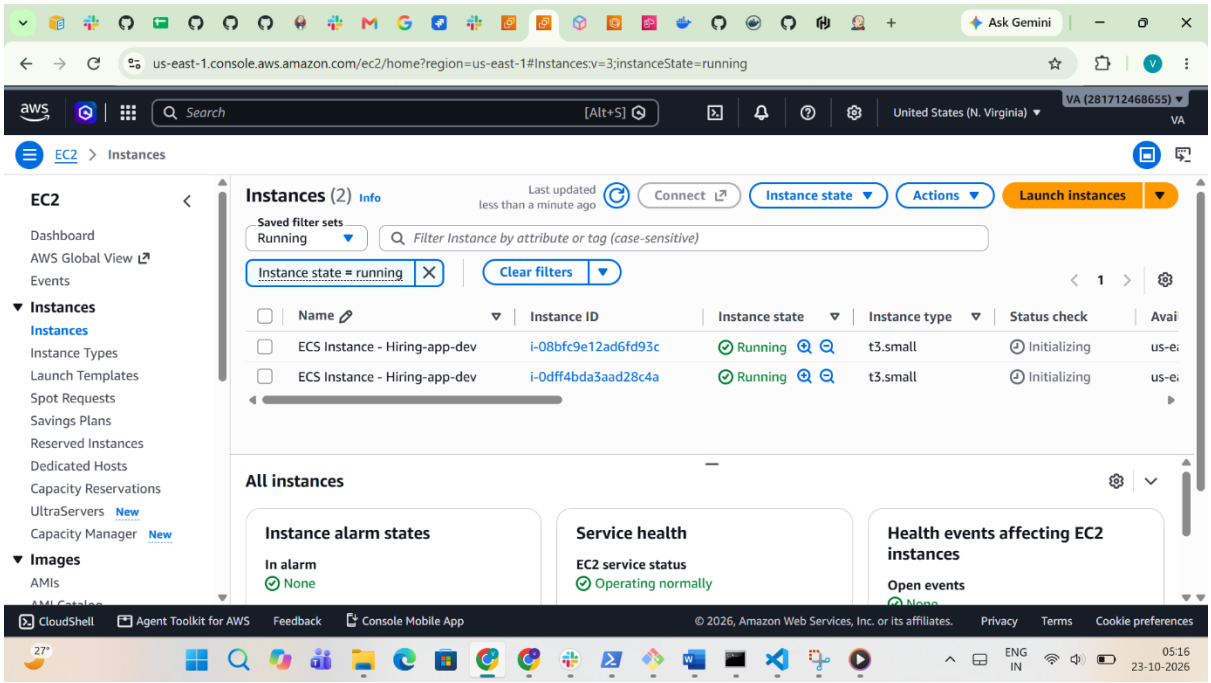
Use the below image from deployment:
sabair0509/hiring-app:works

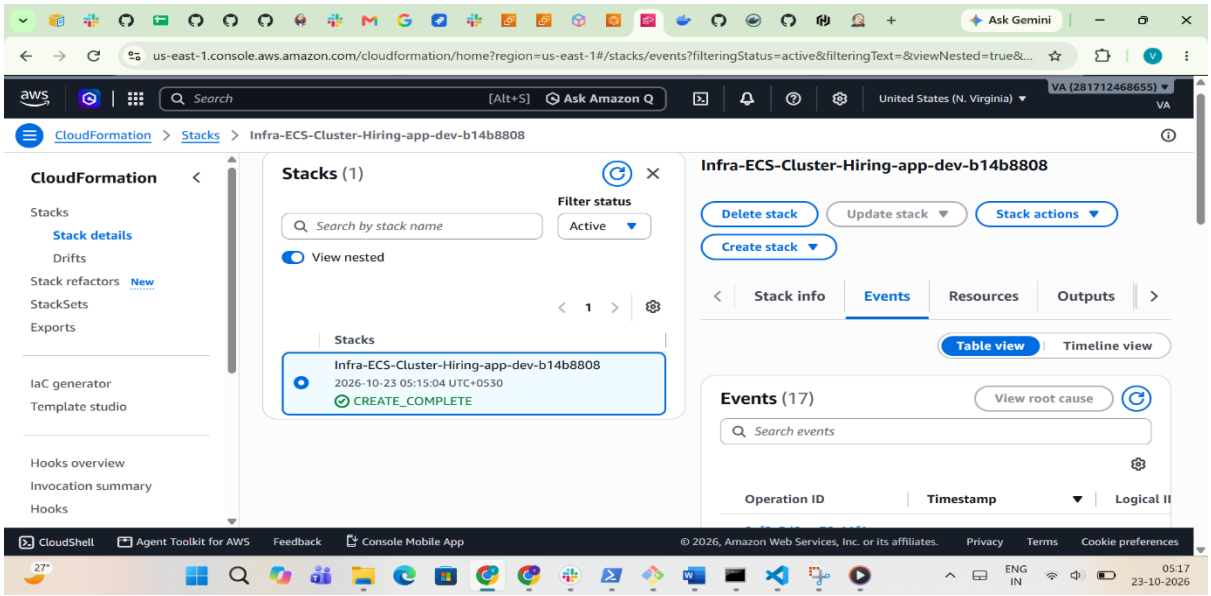
Implementation Steps

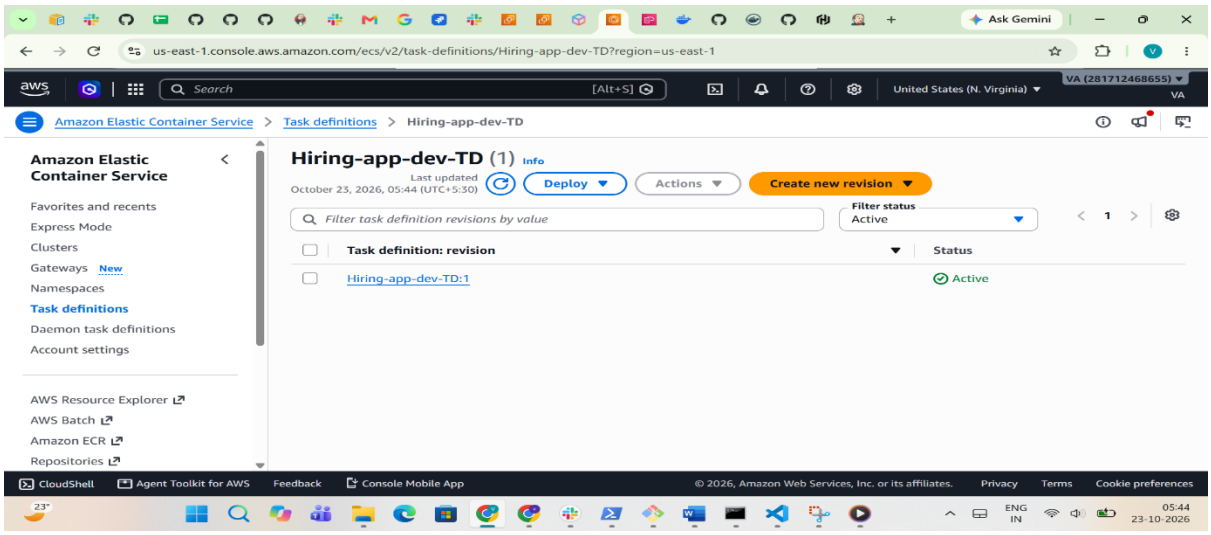
1. Logged in to the AWS Management Console and selected the US East (N. Virginia) us-east-1 region.
2. Selected the default VPC and identified subnets in us-east-1a and us-east-1b.
3. Created an ECS cluster named hiring-app-ha-cluster.
4. Selected Fargate and Self-managed instances and configured a new Auto Scaling group using the On-demand provisioning model.
5. Selected the Amazon ECS-Optimized Amazon Linux 2023 AMI and configured EC2 instances across both Availability Zones.
6. Configured the ECS instance IAM role and cluster registration settings.
7. Created a task definition named hiring-app-task using the image sabair0509/hiring-app:works.
8. Configured the task definition to use EC2 launch capacity and bridge network mode.
9. Configured container port 8080 and host port 0 for dynamic port mapping.
10. Created an internet-facing Application Load Balancer named hiring-app-alb across both Availability Zones.
11. Created a target group named hiring-app-tg with target type Instances and configured the application health check.
12. Configured the ALB listener on HTTP port 80 to forward traffic to the target group.
13. Configured security groups to allow HTTP traffic through the ALB and application traffic from the ALB to the ECS instances.
14. Created an ECS service named hiring-app-ha-service using the hiring-app-task task definition.

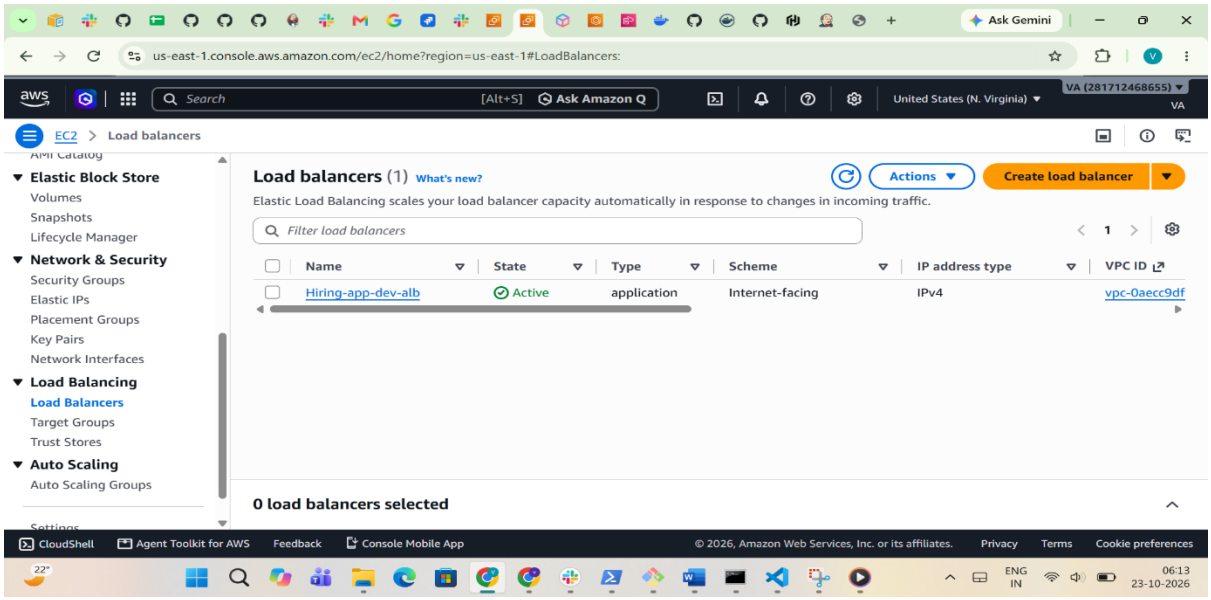
15. Set the desired task count to 6 and configured task placement to spread tasks across both Availability Zones.
16. Attached the ALB and target group to the ECS service using container port 8080.
17. Configured ECS service auto scaling with a minimum of 6 tasks and a maximum of 12 tasks.
18. Ensured EC2 Auto Scaling capacity could support the required task count and replace failed instances.
19. Verified that at least six tasks were running and distributed across both Availability Zones.
20. Verified that the ALB target group showed healthy targets with dynamically assigned host ports.
21. Accessed the application using the ALB DNS name.

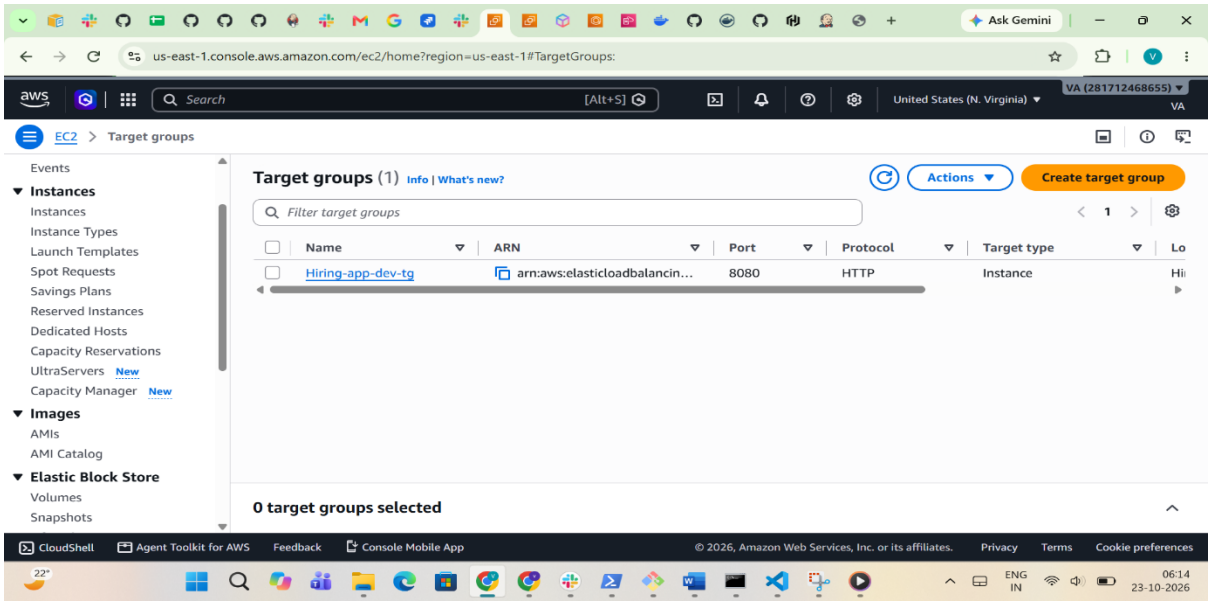


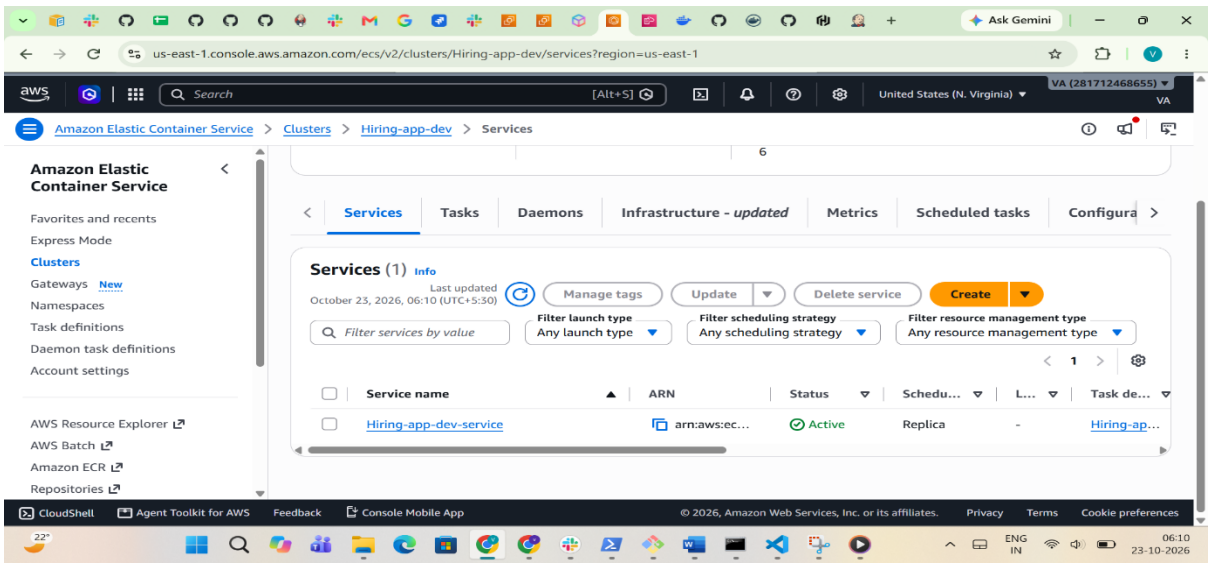

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:v=3;instanceState=running
EC2 > Instances
Instances (2) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Running
Filter Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
ECS Instance - Hiring-app-dev
i-08bfc9e12ad6fd93c
Running
t3.small
Initializing
ECS Instance - Hiring-app-dev
i-0dff4bda3aad28c4a
Running
t3.small
Initializing
All instances
Instance alarm states
In alarm
None
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events

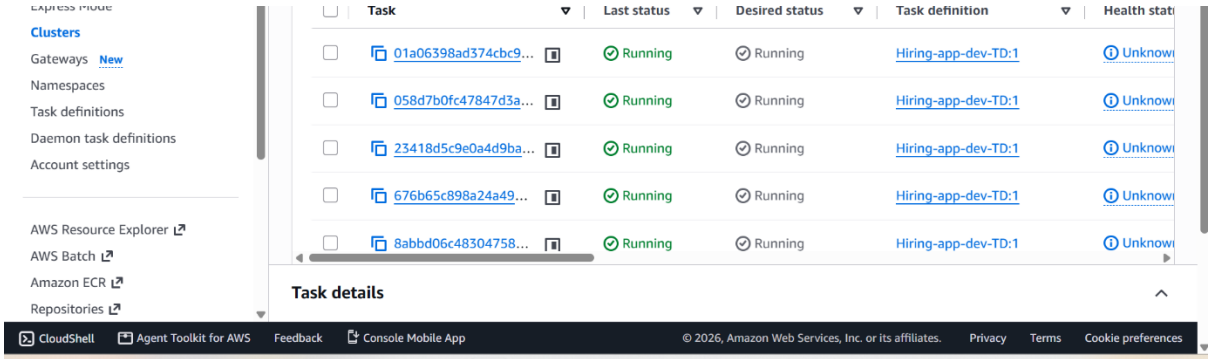

us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/events?filteringStatus=active&filteringText=&viewNested=true&...
CloudFormation > Stacks > Infra-ECS-Cluster-Hiring-app-dev-b14b8808
Stacks (1)
Filter status
Active
View nested
Infra-ECS-Cluster-Hiring-app-dev-b14b8808
2026-10-23 05:15:04 UTC+0530
CREATE_COMPLETE
Infra-ECS-Cluster-Hiring-app-dev-b14b8808
Delete stack
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Table view
Timeline view
Events (17)
View root cause
Search events
Operation ID
Timestamp


us-east-1.console.aws.amazon.com/ecs/v2/task-definitions/Hiring-app-dev-TD?region=us-east-1
Amazon Elastic Container Service > Task definitions > Hiring-app-dev-TD
Hiring-app-dev-TD (1) Info
Last updated October 23, 2026, 05:44 (UTC+5:30)
Deploy
Actions
Create new revision
Filter task definition revisions by value
Filter status
Active
Task definition: revision
Status
Hiring-app-dev-TD:1
Active


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancers:
EC2 > Load balancers
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Load balancers (1) What's new?
Actions
Create load balancer
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.
Filter load balancers
Name
State
Type
Scheme
IP address type
VPC ID
Hiring-app-dev-alb
Active
application
Internet-facing
IPv4
vpc-0aecc9df
0 load balancers selected


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#TargetGroups:
EC2 > Target groups
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Target groups (1) Info | What's new?
Actions
Create target group
Filter target groups
Name
ARN
Port
Protocol
Target type
Hiring-app-dev-tg
arn:aws:elasticloadbalancin...
8080
HTTP
Instance
0 target groups selected


us-east-1.console.aws.amazon.com/ecs/v2/clusters/Hiring-app-dev/services?region=us-east-1
Amazon Elastic Container Service > Clusters > Hiring-app-dev > Services
Amazon Elastic Container Service
Favorites and recents
Express Mode
Clusters
Gateways New
Namespaces
Task definitions
Daemon task definitions
Account settings
AWS Resource Explorer
AWS Batch
Amazon ECR
Repositories
Services
Tasks
Daemons
Infrastructure - updated
Metrics
Scheduled tasks
Services (1) Info
Last updated October 23, 2026, 06:10 (UTC+5:30)
Manage tags
Update
Delete service
Create
Filter services by value
Filter launch type
Any launch type
Filter scheduling strategy
Any scheduling strategy
Filter resource management type
Any resource management type
Service name
ARN
Status
Hiring-app-dev-service
Active
Replica





Welcome to Techie Horizon